
Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

Abstract

Conformal prediction provides distribution-free coverage guarantees, but these degrade under distribution shift—and practitioners currently lack tools to anticipate *which* deployed models will fail before deployment. We propose SHAP concentration—the fraction of feature importance concentrated in the top feature—as a **pre-deployment diagnostic** for conformal prediction vulnerability. Using COVID-19 as a case study across 8 supply chain tasks experiencing varying degrees of feature turnover, we find coverage drops ranging from 0.1% to 77% (all paired $p \leq 0.005$, 50 seeds). SHAP concentration, computed *before* observing test data, correlates with which tasks fail: $\rho = 0.833$, $p = 0.010$ within supply chain ($n = 8$); $\rho = 0.691$, $p = 0.019$ across 3 domains ($n = 11$ including clinical trials and motorsport). Binary classification tasks are structurally protected against this failure mode, clarifying the diagnostic’s scope to multiclass settings. We provide mechanistic intuition: when a model concentrates importance on a single out-of-distribution feature, test conformity scores stochastically dominate calibration scores, breaking the exchangeability assumption. In contrast, post-hoc diagnostics (prediction entropy, calibration error) can only detect failures *after* deployment. Quarterly retraining partially mitigates vulnerable task coverage loss (+19 pp, $p = 0.04$) while robust tasks need no intervention. Adaptive Conformal Inference recovers coverage for vulnerable tasks but inflates prediction sets to 30–82% of classes for those needing intervention (Useful to Vacuous); robust tasks need no ACI. We formalize a decision framework: compute SHAP concentration pre-deployment; retrain quarterly if vulnerable; skip retraining if robust.

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience identical temporal shift yet exhibit coverage drops ranging from negligible (<1%) to catastrophic (>70%).

Our contributions:

1. **Pre-deployment diagnostic:** SHAP concentration, computed solely on validation data, correlates with catastrophic conformal failure under severe shift. Within supply chain: $\rho = 0.833$, $p = 0.010$ ($n = 8$). Across 3 domains (supply chain, clinical trials, motorsport): $\rho = 0.691$, $p = 0.019$ ($n = 11$). SHAP concentration achieves stronger correlation than alternative pre-deployment diagnostics—native feature importance ($\rho = 0.667$, $p = 0.071$) and ensemble disagreement ($\rho = 0.452$, $p = 0.260$)—though the difference is not formally testable at $n = 8$. Binary APS exhibits a structural ceiling effect; concentration is diagnostic specifically for multiclass settings.
2. **Mechanistic intuition:** We provide a heuristic argument that single-feature dependence on an OOD feature causes test conformity scores to stochastically dominate calibration scores, breaking exchangeability (Section 4). We verify this directly: for catastrophic tasks, test conformity scores stochastically dominate

calibration scores (KS = 0.68–0.96, all $p < 10^{-10}$; Figure 1).

3. **Quantification with statistical rigor:** Coverage drops across 8 tasks range from 0.1% to 77.1%, all with paired $p \leq 0.005$ (Wilcoxon, 50 seeds). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though significance is not maintained for 2 of 8 jackknife samples due to reduced power at $n = 7$.
4. **Actionable framework:** Quarterly retraining partially mitigates vulnerable task coverage loss by +19 pp ($p = 0.04$); robust tasks (99.8% coverage) need no intervention. The framework’s concentration threshold is robust across 30–45% (Recall = 1.0; Appendix D).
5. **Binary ceiling effect:** Binary APS prediction sets ($\{0\}$, $\{1\}$, or $\{0, 1\}$) are structurally protected against the coverage degradation that concentration detects. This boundary condition clarifies when the diagnostic applies (multiclass) and when it does not (binary).

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [Romano et al., 2020] and regression [Romano et al., 2019]. Angelopoulos and Bates [2021] provide a comprehensive tutorial.

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free prediction sets under label shift. Barber et al. [2023] provide a comprehensive treatment beyond exchangeability. These works characterize degradation theoretically; we complement them with an empirical diagnostic that identifies *which* tasks will fail before deployment.

Adaptive Methods. Gibbs and Candès [2021] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series and Feldman et al. [2023] for online risk control. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show robust methods often fail under temporal shift. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Existing shift-detection methods (e.g., prediction entropy, calibration error) are *post-hoc*. Our contribution is a *pre-deployment* diagnostic using feature importance structure, requiring no test data.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017] to model reliability assessment, extending the use of feature attribution from debugging [Adebayo et al., 2018] to prospective failure diagnosis.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Fey et al., 2023], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded.

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

Data separation protocol. SHAP concentration and the 40% threshold are computed exclusively on validation-time data. The test set is used only for final evaluation and never informs diagnostic thresholds. Task-level severity labels (SEV/ROB) used for threshold evaluation are defined by observed test coverage drops; the threshold sensitivity analysis in Appendix D is therefore post-hoc validation, not prospective assessment.

3.2 CONFORMAL PREDICTION SETUP

We use Adaptive Prediction Sets (APS) [Romano et al., 2020] with $\alpha = 0.1$ (90% target coverage):

1. Train LightGBM classifier with default hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

We train 50 independent models with seeds 42–91. For each seed s : train model M_s , split validation 50/50 into calibration $\mathcal{D}_{\text{cal}}^s$ and evaluation $\mathcal{D}_{\text{eval}}^s$, calibrate CP_s on $\mathcal{D}_{\text{cal}}^s$, evaluate on both $\mathcal{D}_{\text{eval}}^s$ and test. We report mean, 95% confidence intervals (t -distribution), and paired Wilcoxon tests (seed-level). This is *not* ensemble prediction—each seed is an independent experimental trial providing paired comparisons.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(f) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature f in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where ϕ_j is the mean absolute SHAP value of the j -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 2), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). SHAP concentration is the only diagnostic available before deployment.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? We compared top-1 concentration against alternatives: top-2, top-3, Herfindahl-Hirschman Index (HHI), and feature importance entropy. Only top-1 concentration achieves significant correlation with coverage drop ($\rho = 0.833$, $p = 0.010$); all alternatives are non-significant ($p > 0.10$). The discriminating power of top-1 arises because sales-office has high top-2/3 concentration (77–85%) but near-zero coverage drop—the diagnostic signal is specifically whether a *single* dominant feature drives predictions, not overall feature spread.

4 INTUITION: WHY CONCENTRATION CORRELATES WITH FAILURE

We provide intuition for why SHAP concentration correlates with conformal failure under severe feature shift. The following argument is heuristic; formal analysis under explicit distributional assumptions is left for future work.

Setup. Consider a model f trained on features $(x_1, \dots, x_p)$, calibrated with conformal predictor CP at level α . For APS, the conformity score is $s(x, y) = \sum_{j=1}^k \hat{f}_{\pi_j}(x)$, where π orders classes by decreasing $\hat{f}_j(x)$ and $\pi_k = y$ (i.e., the cumulative probability mass from the most likely class until the true label is included). The conformal guarantee requires that calibration and test scores be exchangeable.

Claim (heuristic). If (i) the model concentrates importance on feature x_1 (i.e., $C \gg 0$), and (ii) x_1 undergoes complete distribution shift ($J(x_1) \approx 0$), then the conformity scores on the test set stochastically dominate the calibration scores (i.e., test scores are systematically larger), and the coverage guarantee breaks.

Argument. When C is high, the model’s predictions are primarily determined by x_1 . Under complete shift, test-time values of x_1 are entirely unseen during training. The model maps these to arbitrary predictions (effectively random with respect to the true label distribution), producing conformity scores s_{test} that are systematically higher than s_{cal} . Formally:

$$P(s_{\text{test}} \leq t) \leq P(s_{\text{cal}} \leq t), \quad \forall t \quad (3)$$

This stochastic dominance means the conformal threshold $\hat{q}$ (calibrated to achieve $1 - \alpha$ coverage on calibration data) becomes too small for test data, producing prediction sets that systematically exclude the true label.

Diversification protects. When importance is distributed across p features (low C), the effect of any single OOD feature is attenuated. Even if all features shift, the model’s score distribution depends on a “portfolio” of features whose individual effects partially cancel under resampling. This is analogous to portfolio diversification in finance: concentrated bets amplify risk; diversified exposure dampens it.

Empirical verification. Figure 2 is consistent with this argument: catastrophic tasks show concentrated importance explosion ($4.5\times$) in the top feature, while robust tasks redistribute importance across features ($10\text{--}20\times$ increases distributed). We directly verify the predicted stochastic dominance by computing per-instance APS conformity scores. For the most catastrophic task (sales-shipcond, 71.6% drop), calibration and test score distributions differ dramatically ($\text{KS} = 0.956$, $p < 10^{-10}$): calibration scores spread across $[0.2, 1.0]$ (mean = 0.54) while test scores concentrate near 1.0 (mean = 0.98), confirming test scores stochastically dominate calibration scores (Figure 1). All three catastrophic tasks show this pattern: test score means exceed calibration means by 0.02–0.44. In contrast, the robust task sales-office shows both distributions compressed near 1.0 (means 0.996 vs. 1.000). Full KS statistics for all 8 tasks in Appendix H.

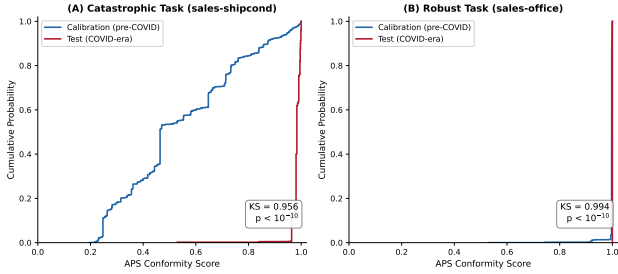


Figure 1: **Conformity Score Stochastic Dominance.** (A) Catastrophic task (sales-shipcond): test scores concentrate near 1.0 while calibration scores spread across $[0.2, 1.0]$ ($KS = 0.956$). (B) Robust task (sales-office): both distributions compressed near 1.0.

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing identical COVID-19 temporal shift, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 1). All coverage drops are statistically significant (paired Wilcoxon $p \leq 0.005$, 95% CIs from the t -distribution), yet range from 0.1% to 77.1%.

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 2).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Among 8 tasks with severe feature turnover, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 3, Figure 3). Bootstrap 95% CI: $[0.29, 1.00]$ (10,000 resamples). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 2 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated im-

portance explosion ($4.5\times$ increase). Robust tasks: importance redistributes across features with ranking reshuffle ($10\text{--}20\times$ increases distributed). This matches the heuristic prediction in Section 4.

False positives. Two tasks with concentration $>40\%$ remain robust: sales-office (secondary feature SALESORG has Jaccard = 0.61) and item-shippoint (high model variance). These motivate the protective-factor check in our framework (Section 7).

Task independence. Intraclass correlation analysis confirms task identity explains the majority of seed-level variance in coverage drops (ICC = 0.675, 95% CI $[0.47, 0.90]$; effective $n_{\text{eff}} = 11.7$). Per-task ICC(val,test) is negative for all 8 tasks, indicating paired tests are conservative. Details in Appendix G.

Cross-domain extension. Extending to 11 tasks across 3 domains (supply chain, clinical trials, motorsport; Section 6.3) yields $\rho = 0.691$, $p = 0.019$. The attenuation from 0.833 to 0.691 is entirely attributable to the binary ceiling effect: binary APS tasks cluster near zero drop regardless of concentration.

This association is observational; prospective validation on held-out domains is needed before the diagnostic can be used for forward-looking prediction.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 4).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds (seed 42). Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as confound. $\log(\text{num_classes})$ alone correlates with coverage drop ($\rho = 0.743$, $p = 0.035$). However, partial correlations reveal an asymmetry: concentration retains $\rho_{\text{partial}} = 0.629$ after controlling for $\log(\text{num_classes})$, while $\log(\text{num_classes})$ drops to $\rho_{\text{partial}} = 0.334$ after controlling for concentration. Neither achieves significance at $n = 8$, but the asymmetry suggests concentration captures information beyond class count.

We additionally compare against two *post-hoc* baselines

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t -distribution. p -values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	CI	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	462	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	135	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	70	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB*
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV > 50%), classified by median.

Table 2: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val — mean test coverage (50 seeds).

Table 4: Pre-Deployment Diagnostic Comparison ($n = 8$). Spearman ρ with 50-seed mean coverage drop. ρ_{partial} : partial Spearman controlling for the other variable in the SHAP-class-count pair.

Diagnostic	ρ	p	ρ_{partial}	p_{partial}
SHAP concentration	0.833	0.010	0.629	0.131
log(num_classes)	0.743	0.035	0.334	0.464
Native FI concentration	0.667	0.071	—	—
Ensemble disagreement	0.452	0.260	—	—

Table 3: Stratified Correlation Analysis.

Group	n	Spearman ρ	p -value	Boot. 95% CI
Multiclass (SALT)	8	0.833	0.010	[0.29, 1.00]
COVID-era	9	0.883	0.002	[0.39, 1.00]
Combined (3 domains)	11	0.691	0.019	[0.08, 1.00]

LOO stability (SALT): $\rho \in [0.75, 0.96]$; significance maintained in 6/8 jackknife samples ($p < 0.05$); 2 samples at $p = 0.052$ due to reduced power at $n = 7$. Combined correlation attenuated by binary ceiling effect (Section 6.3).

computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

Expected Calibration Error change: $\Delta \text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6 EXTENDED EXPERIMENTS

6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

We test ACI [Gibbs and Candès, 2021] on all 8 tasks (7 with 10 seeds, item-incoterms as 3-seed pilot; $\gamma = 0.01$ following the original recommendation; results for $\gamma \in \{0.001, 0.05\}$ are qualitatively similar). ACI updates the miscoverage level online: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. Table 5 reveals a clear contrast between task categories.

Severe tasks: ACI provides large coverage gains (+48–72pp, all $p < 0.01$) but at varying cost to informativeness (Utility column). For sales-group (462 classes), ACI expands sets to 82% of all classes—formally “Vacuous,” predicting nearly every class is possible. Sales-payterms is “Marginal” (40% of classes). Only sales-shipcond retains useful sets (30% of classes) despite the severe shift.

Robust tasks: ACI provides marginal, mostly non-significant improvement (+1.5–10pp). Sales-office is instructive: at 99.9% standard coverage, ACI slightly *reduces* coverage (−1.0pp, n.s.) as the adaptive mechanism overshoots toward the 90% target. All robust tasks maintain “Useful” prediction sets—they already achieve near-target coverage without ACI.

Synthesis: Across all 8 tasks, ACI improves coverage for vulnerable tasks but utility depends on set-size growth; vacu-

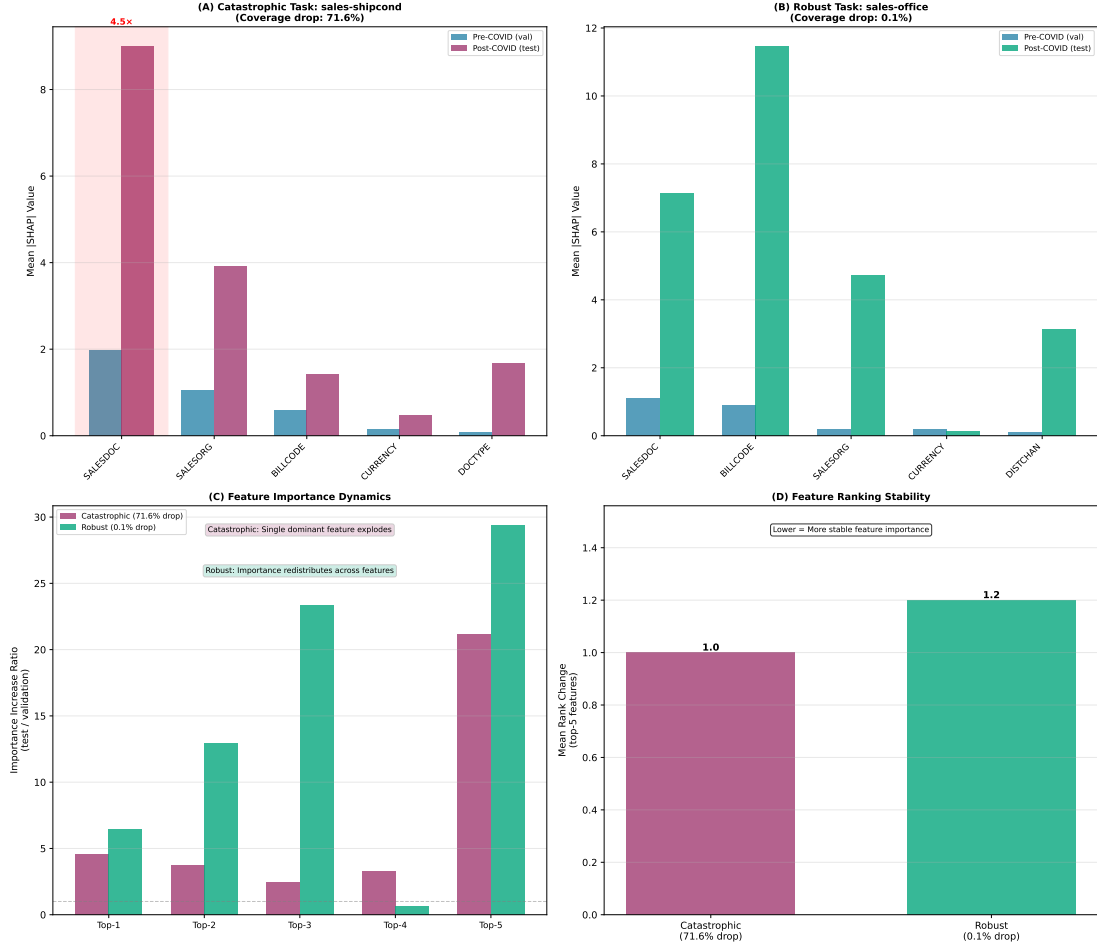


Figure 2: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes $4.5\times$ while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite $\sim 0\%$ Jaccard similarity, coverage drops range from negligible to catastrophic.

ity appears in high-class-cardinality settings (s-group: 462 classes $\rightarrow$ 82% Vacuous). For robust tasks, ACI is unnecessary or counterproductive.

Deployability metric: We define the *usable-set rate* as $\text{Size}/|\mathcal{Y}|$, the mean prediction set size divided by the number of classes. Values below 40% indicate actionable predictions (“Useful”); above 60% indicates near-vacuous output. Of the 3 severe tasks where ACI is needed, only 1 (s-shipcond, 30%) produces deployable predictions. This reinforces our diagnostic: identify vulnerability *before* deployment rather than relying on ACI to fix it *after*.

6.2 RETRAINING ANALYSIS

We test four retraining frequencies on catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020).

Catastrophic task (Table 6, Figure 4): Quarterly retrain-

ing achieves the highest mean coverage (41.1%), a statistically significant improvement of +18.9 pp over no retraining (Wilcoxon $p = 0.04$). Monthly retraining shows higher variance and occasional collapse (min: 0.6%). Quarterly offers the best cost-effectiveness: 3 retrains/year achieves 85% of maximum improvement.

Robust task: Coverage remains 99.8–99.9% regardless of retraining frequency, confirming distributed feature importance eliminates need for retraining.

6.3 CROSS-DOMAIN VALIDATION

To test whether SHAP concentration generalizes beyond a single database, we compute APS coverage drops (50 seeds) and SHAP concentration for binary classification tasks across two additional domains (Table 7).

Combined correlation. Across all 11 tasks from 3 domains, Spearman $\rho = 0.691$, $p = 0.019$ (permutation

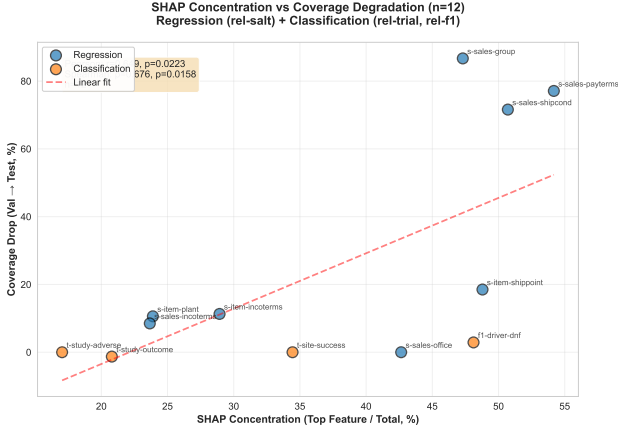


Figure 3: **SHAP Concentration Correlates with Coverage Degradation.** Spearman $\rho = 0.833$, $p = 0.010$ for 8 severe-shift tasks (dark). Lighter points: 3 binary/cross-domain tasks (ceiling effect; Section 6.3). The 40% threshold (dashed) is set on validation data only.

$p = 0.023$; bootstrap 95% CI $[0.08, 0.97]$). LOO stability: $\rho \in [0.59, 0.79]$, 7/11 jackknife samples significant at $p < 0.05$.

Binary ceiling effect. The three binary tasks (study-outcome, driver-dnf, driver-top3) show near-zero coverage drops (-1.3% , $+2.9\%$, $+1.2\%$) despite SHAP concentrations spanning 20.8% to 48.1%. This reflects a *structural* property of binary APS: prediction sets are restricted to $\{0\}$, $\{1\}$, or $\{0, 1\}$, leaving little room for coverage degradation regardless of shift severity. Mean binary drop = 0.9% vs. multiclass = 33.6% (Mann-Whitney $p = 0.024$). Adding binary tasks weakens the overall correlation from $\rho = 0.833$ to 0.691 —not because the diagnostic fails, but because binary APS is structurally protected against the failure mode that concentration detects.

COVID-era subset. Restricting to the 9 COVID-era tasks (8 SALT + study-outcome) yields $\rho = 0.883$, $p = 0.002$ (bootstrap 95% CI $[0.39, 1.00]$; all 9 LOO samples significant). This stronger result suggests SHAP concentration is particularly diagnostic under severe, event-driven shift.

Interpretation. SHAP concentration correlates with multiclass APS failure across domains when severe feature shift is present. Binary classification tasks exhibit a ceiling effect that limits APS vulnerability regardless of feature importance structure. This boundary condition is itself informative for practitioners: binary conformal predictors are structurally more robust to the failure mode we identify.

6.4 PLACEBO TEST

To establish that COVID-19 represents a unique shift, we use pre-COVID data: train on 2018, validate on 2019-H1,

Table 5: ACI Across Task Categories ($\gamma = 0.01$, $\alpha = 0.1$). 10 seeds except † item-incoterms (3-seed pilot). Δ = paired coverage improvement. p from Wilcoxon signed-rank. Utility: Useful ($<40\%$), Marginal ($40\text{--}60\%$), Vacuous ($>60\%$).

Task	Std Cov.	ACI Cov.	Δ	Size/#cl	p	Utility
<i>Severe tasks</i>						
s-shipcond	33.0%	80.5%	+47.5pp	30%	0.004	Useful
s-group	10.4%	82.2%	+71.8pp	82%	0.002	Vacuou
s-payterms	11.1%	71.1%	+60.0pp	40%	0.005	Margina
<i>Robust tasks</i>						
i-plant	79.9%	89.9%	+10.1pp	32%	0.005	Useful
i-shippoint	82.1%	89.9%	+7.9pp	34%	0.615	Useful
s-incoterms	88.5%	90.0%	+1.5pp	35%	0.812	Useful
i-incoterms †	88.9%	90.0%	+1.2pp	$\sim 35\%$	n.s.	Useful
s-office	99.9%	98.9%	$-1.0pp$	5%	0.979	Useful

Standard coverage uses 10-seed subset (differs slightly from 50-seed Table 1). † 3 seeds only (compute-intensive); high model variance (Std: 76–95%), ACI stabilizes at 90%.

Table 6: Retraining Impact on Catastrophic Task (Sales-Shipcond).

Frequency	Mean Cov.	Min Cov.	p vs none
No retrain	22.2%	10.8%	—
Bi-annual	27.0%	11.4%	0.04
Quarterly	41.1%	20.7%	0.04
Monthly	32.0%	0.6%	<0.05

test on 2019-H2 (Table 8 in Appendix). COVID causes $6\text{--}140\times$ more degradation than normal temporal drift for vulnerable tasks. This confirms that the observed coverage collapse is attributable to COVID-19 disruption rather than routine temporal drift, and that the SHAP concentration diagnostic identifies vulnerability specific to the severe shift regime.

7 DECISION FRAMEWORK

The following protocol, empirically motivated by our analysis, guides practitioners deploying conformal prediction in non-stationary settings:

1. **Compute SHAP concentration** $C = \phi_1 / \sum_j \phi_j$ on validation data (Eq. 2)
2. **If $C \leq 40\%$:** Task is **robust** (distributed importance protects against shift)
3. **If $C > 40\%$:** Check for protective factors—secondary features with train-validation Jaccard > 0.5 and importance $> 15\%$ (validation overlap serves as a pre-deployment proxy for deployment-time feature stability)

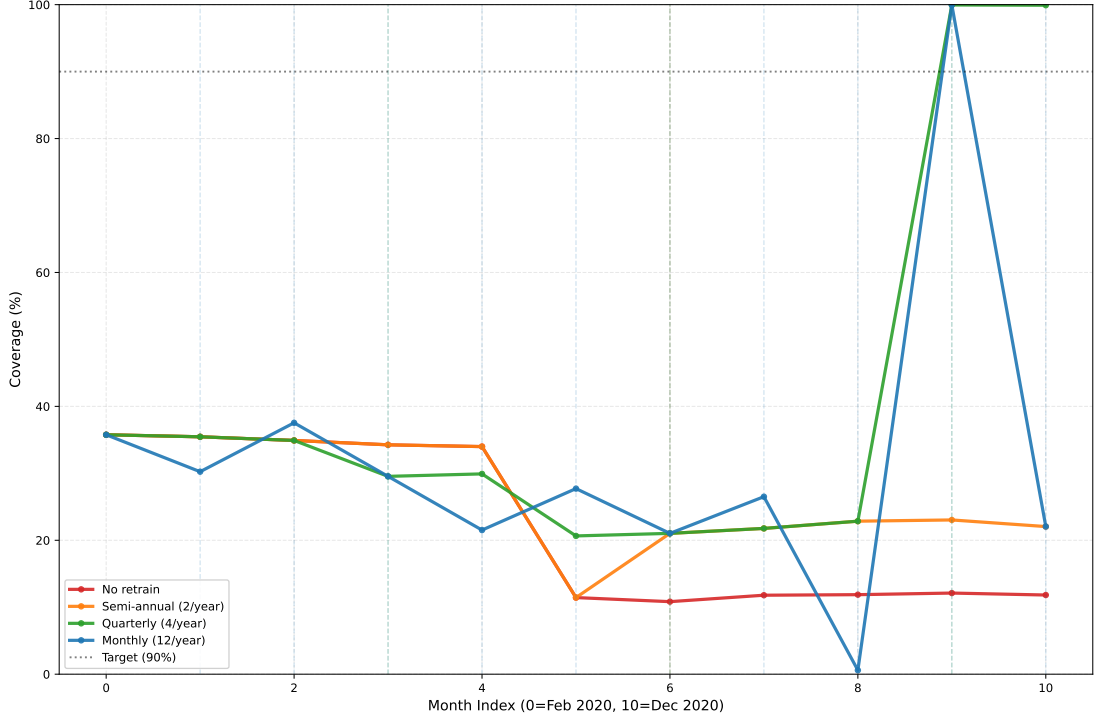


Figure 4: **Retraining partially mitigates coverage loss for catastrophic tasks.** Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly (blue) shows high variance. No retraining (red) exhibits severe degradation post-COVID.

- Protective factors exist → **robust**
- Otherwise → **vulnerable**: implement quarterly retraining

4. **For robust tasks**: Skip retraining (saves computational cost)
5. **Monitor**: Track empirical coverage as early warning

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 10. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. Here we define “at-risk” as coverage drop $>15\%$ (encompassing both SEV and moderate-drop tasks), broadening the detection target beyond the extreme SEV category ($>50\%$ drop, Table 1) to capture any non-trivial failure. Applying the 40% threshold without re-tuning to the full $n = 11$ cross-domain set (Table 7) yields Recall = 1.0, F1 = 0.80 (or F1 = 0.67 under the stricter $>50\%$ SEV definition). The binary tasks do not violate the threshold—study-outcome (20.8%, below threshold) is correctly classified as robust, while driver-dnf (48.1%, above threshold) is a false positive only because binary APS is structurally protected. The 45% threshold achieves F1

= 0.89 (or 0.75 under $>50\%$) at $n = 11$. The sensitivity analysis in Appendix D reports performance across the 30–50% range.

8 DISCUSSION

Confidently Wrong Models. Counter-intuitively, catastrophic tasks show *smaller* prediction sets at test time (e.g., sales-shipcond: $7.0 \rightarrow 3.0$). The model concentrates probability on a small number of wrong classes rather than spreading uncertainty. This is mechanistically consistent with our theory: when the model depends on a single OOD feature, it maps unseen values to arbitrary but *concentrated* probability distributions. Appendix F confirms that prediction entropy *decreases* for catastrophic tasks—the opposite of what entropy-based monitoring would flag.

High Variance as Warning. Tasks with CV $> 50\%$ across seeds are in knife-edge regimes. This is itself an early warning that conformal calibration will be unreliable.

Alternative Conformal Methods. Mondrian conformal prediction [Vovk et al., 2005] calibrates per-class, but when the dominant feature undergoes complete shift (Jaccard ≈ 0), test-time values have no calibration counterpart. Weighted conformal prediction [Tibshirani et al., 2019] faces the same obstacle: likelihood ratio weights require distributional over-

Table 7: Cross-Domain Validation: SHAP Concentration and APS Coverage (50 Seeds). Supply chain tasks from primary analysis; clinical trials and motorsports are independent datasets with separate temporal splits.

Dataset / Task	CI	Conc. (%)	Drop (%)	Domain	Shift
<i>Supply chain (rel-salt, COVID Feb–Jul 2020)</i>					
s-payterms	135	54.2	77.1	SC	COVID
s-shipcond	45	50.7	71.6	SC	COVID
s-group	462	47.3	71.2	SC	COVID
i-shippoint	70	48.8	18.5	SC	COVID
s-office	25	42.6	0.1	SC	COVID
i-incoterms	13	28.9	11.3	SC	COVID
i-plant	35	23.9	10.6	SC	COVID
s-incoterms	13	23.7	8.5	SC	COVID
<i>Clinical trials (rel-trial, COVID Jan 2020–Jan 2021)</i>					
study-outcome	2	20.8	−1.3	CT	COVID
<i>Motorsport (rel-fl, pre-COVID 2005–2010)</i>					
driver-dnf	2	48.1	2.9	MS	None
driver-top3	2	36.8	1.2	MS	None

lap. SHAP concentration describes the *fixed model’s* learned dependence structure (computed on validation data), not a property of the test distribution, making it applicable precisely when these methods cannot be.

Limitations.

Generalizability. (1) Cross-domain validation extends to 11 tasks across 3 domains, but the core SALT analysis (8 tasks) shares a single database, temporal split, and model class. The effective independence of SALT observations is limited by shared structure; cross-domain tasks partially mitigate this ($\rho = 0.691$, $p = 0.019$ at $n = 11$). (2) The 40% threshold is exploratory, derived from $n = 8$ multiclass tasks. Cross-domain transfer (Recall = 1.0 at $n = 11$) is encouraging but insufficient for deployment recommendations; practitioners should calibrate thresholds on held-out domains. (3) Binary APS tasks exhibit a structural ceiling effect (Section 6.3) that limits both the failure mode and the diagnostic’s applicability; the concentration diagnostic is validated for multiclass settings only.

Methodology. (4) The 50 random seeds share identical training and test data, varying only in model randomness and calibration splits. ICC analysis (Appendix G) confirms task identity explains most variance (ICC = 0.675, effective $n_{\text{eff}} = 11.7$), and per-task negative ICC(val,test) indicates paired tests are conservative. After Holm–Bonferroni correction for 5 concentration metrics tested, the primary result reaches $p_{\text{adj}} = 0.050$ (Appendix G). (5) Among pre-deployment diagnostics (Section 5.4), SHAP concentration achieves stronger correlation than native feature importance ($\rho = 0.667$, $p = 0.071$) and ensemble disagreement ($\rho = 0.452$, $p = 0.260$), though the difference between dependent correlations is not formally testable at $n = 8$.

Scope. (6) The diagnostic is model-specific: RandomForest trained on the same data shows different concentration profiles and the correlation does not replicate ($\rho = 0.30$, $p = 0.47$; Appendix I). This is expected—SHAP concentration measures a model’s learned dependence structure, not a data-level property—but means the diagnostic must be computed for the deployed model class. (7) The protective-factor thresholds (Jaccard > 0.5, importance > 15%) are preliminary heuristics validated on 1 example. (8) The intuition in Section 4 is heuristic; formal analysis under explicit distributional assumptions remains open.

CONCLUSION

We propose SHAP concentration as a pre-deployment diagnostic for conformal prediction vulnerability under distribution shift. Using COVID-19 as a case study across 11 tasks in 3 domains: (1) Coverage drops range from negligible (<1%) to catastrophic (>70%) under identical shift, all statistically significant ($p \leq 0.005$); (2) SHAP concentration correlates with which multiclass tasks fail ($\rho = 0.833$ within supply chain, $\rho = 0.691$ across 3 domains); (3) we provide mechanistic intuition via conformity score stochastic dominance, verified empirically (KS = 0.68–0.96 for catastrophic tasks); (4) binary APS is structurally protected against this failure mode regardless of concentration; (5) quarterly retraining partially mitigates vulnerable coverage loss by +19 pp ($p = 0.04$); (6) ACI recovers coverage for vulnerable tasks but inflates sets to 30–82% of classes for those needing intervention; robust tasks need no ACI.

The diagnostic is pre-deployment (validation data only), actionable (retrain/skip decision), and stable (threshold effective across 30–45%).

More broadly, our results suggest that the *structure* of learned feature importance—not just the presence of distribution shift—determines whether conformal guarantees hold in practice. This opens a direction for pre-deployment model auditing based on importance concentration analysis.

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Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

A REPRODUCIBILITY DETAILS

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with default LightGBM settings: objective=multiclass, boosting=gbdt, num_leaves=31, learning_rate=0.05, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5, num_boost_round=500, early_stopping_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage: $(1 - \alpha) = 0.9$ (90%). Calibration: random 50% split of validation set. Quantile: $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation (pre-COVID) data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: ~3–4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

B PLACEBO TEST DETAILS

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 1 exhibit extreme model variance ($CV > 50\%$):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.

Table 8: Placebo Test: Pre-COVID vs COVID Degradation (50-seed mean drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

D THRESHOLD SENSITIVITY ANALYSIS

Table 9 shows the decision framework’s classification performance across threshold values. The threshold is derived from the validation-data concentration gap; task-level ground-truth labels (SEV/ROB) are defined by observed test coverage drops, making this a post-hoc validation.

Table 9: Threshold Sensitivity for SHAP Concentration (Step 2 only, no protective-factor check). Ground-truth task labels (SEV/ROB) are defined by observed test coverage drops; this analysis is post-hoc validation of a validation-derived threshold. Recall = 1.0 across the 30–45% range.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.50	1.00	0.67	3	3	0
30%	0.60	1.00	0.75	3	2	0
35%	0.60	1.00	0.75	3	2	0
40%	0.60	1.00	0.75	3	2	0
45%	0.75	1.00	0.86	3	1	0
50%	1.00	0.67	0.80	2	0	1
55%	0.00	0.00	0.00	0	0	3

After Step 3 (protective-factor check at 40%): Precision = 75%, Recall = 100%, F1 = 86% (in-sample; external validation needed). The single false positive (i-shippoint, concentration 48.8%) has high model variance (median drop 1.2%; most seeds maintain coverage).

E DECISION FRAMEWORK VALIDATION

Table 10: Concentration Check Applied to All Tasks.

Task	Conc. (%)	Step 2	Actual	Note
s-payterms	54.2	VULN	SEV	—
s-shipcond	50.7	VULN	SEV	—
i-shippoint	48.8	VULN	ROB*	High variance [†]
s-group	47.3	VULN	SEV*	—
s-office	42.6	VULN	ROB	Protective [‡]
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—

[†]Median drop 1.2%; most seeds maintain coverage. [‡]SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%.

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Table 11: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-seed ACI experiments). SHAP concentration is computed on validation data only. Δ Entropy and Δ ECE require test-time observations.

Task	Cat	SHAP Conc. [†]	Δ Entropy	Δ ECE
<i>Severe tasks</i>				
s-shipcond	SEV	50.7%	−0.746	+0.408
s-group	SEV	47.3%	−1.274	+0.176
s-payterms	SEV	54.2%	−1.979	+0.629
<i>Robust tasks</i>				
i-plant	ROB	23.9%	+0.181	+0.1314
i-shippoint	ROB	48.8%	+0.469	+0.0727
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

[†]Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

Counter-intuitively, all three catastrophic tasks (coverage drop $> 70\%$) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). SHAP concentration remains the only evaluated metric available before deployment that correctly identifies all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. For coverage drops, $\text{ICC} = 0.675$ (95% CI: [0.47, 0.90]; $F(7, 392) = 104.6$), confirming that task identity explains the majority of seed-level variance. The design effect is 34.1, yielding effective $n_{\text{eff}} = 400/34.1 = 11.7$ —exceeding the 8 task-level observations. For test coverages, $\text{ICC} = 0.720$ (95% CI: [0.52, 0.92]). These values confirm that the $n = 8$ tasks constitute meaningfully independent observations for the Spearman correlation analysis.

Per-task ICC(val,test). For each task, we compute ICC between the 50 validation coverages and 50 test coverages (paired by seed). All 8 per-task ICCs are negative (−0.80 to −0.10), indicating that seeds producing above-average validation coverage do *not* produce above-average test coverage. This means paired Wilcoxon tests are conservative: the effective sample size for within-task paired tests *exceeds* 50, so the reported p -values are, if anything, too large.

Multiple comparisons. We tested 5 concentration metrics (top-1, top-2, top-3, HHI, entropy) against coverage drop. Under Holm–Bonferroni correction for this 5-test family, the adjusted p -value for top-1 concentration is $5 \times 0.010 = 0.050$, at the conventional significance boundary. For the 3 correlation subsets ($n = 8, n = 9, n = 11$), all adjusted p -values remain significant (Holm-corrected $p \leq 0.021$). We note that top-1 was the a priori natural choice (Section 3.6), not one of 5 equally plausible alternatives; the correction is therefore conservative.

H CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 13 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 12: Per-Task ICC Between Validation and Test Coverages (50 Seeds).

Task	ICC(val,test)	Design Effect	Eff. n
s-payterms	−0.80	0.20	250
s-shipcond	−0.77	0.23	220
s-group	−0.72	0.28	177
i-plant	−0.42	0.58	86
i-incoterms	−0.37	0.63	80
s-incoterms	−0.35	0.65	77
s-office	−0.15	0.85	59
i-shippoint	−0.10	0.90	55

Table 13: KS Tests: Calibration vs. Test Conformity Score Distributions (seed=42).

Task	KS	p	Mean _{cal}	Mean _{test}
<i>Catastrophic tasks (test scores > calibration)</i>				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
<i>Robust tasks</i>				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Non-randomized APS scores (conservative coverage guarantee). For catastrophic tasks, test scores shift rightward (stochastic dominance), consistent with the heuristic argument in Section 4. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

I MODEL CLASS SENSITIVITY: RANDOMFOREST

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with `RandomForestClassifier` (500 trees, seed=42, same data pipeline; Table 14).

Table 14: LightGBM vs. RandomForest: SHAP Concentration and Coverage Drop.

Task	LGB Conc	RF Conc	LGB Drop	RF Drop
s-shipcond	50.7%	57.4%	71.6%	15.8%
s-group	47.3%	22.5%	71.2%	50.0%
s-payterms	54.2%	27.9%	77.1%	13.8%
i-shippoint	48.8%	52.1%	18.5%	22.7%
i-plant	23.9%	26.8%	10.6%	15.3%
s-incoterms	23.7%	26.8%	8.5%	2.5%
i-incoterms	28.9%	17.5%	11.3%	2.9%
s-office	42.6%	24.5%	0.1%	0.0%

Result. RF SHAP concentration does *not* correlate with RF coverage drop ($\rho = 0.30$, $p = 0.47$, $n = 8$), in contrast to LightGBM ($\rho = 0.833$, $p = 0.010$). Cross-model concentration correlation is moderate ($\rho = 0.54$, $p = 0.17$).

Interpretation. Two factors explain the discrepancy. First, RF produces different vulnerability profiles: the three LightGBM-catastrophic tasks (71–77% drop) show much smaller RF drops (14–50%), because RF’s bagging produces smoother probability surfaces less sensitive to individual OOD features. Second, RF concentration has a compressed range: 6 of 8 tasks cluster at 22–28%, eliminating the separation between vulnerable and robust tasks that LightGBM exhibits (24–29% vs. 43–54%).

This confirms SHAP concentration is a *model-specific* diagnostic: it measures how a particular model’s learned dependence

structure creates vulnerability, not a universal property of the task or data. Practitioners should compute concentration for their deployed model class.